4.2.15 GPU

The Graphical Processing Unit is the computing core of a graphics card. These chips are designed to perform computations on graphical data much more efficiently than can CPUs. Modern GPUs are more complex than CPUs, if transistor count be considered. One of the fastest GPUs today, the GeForce 8800 GTX, has 754 million transistors, while the fastest CPU, the quad-core Intel Core 2 Quad, has 820 million transistors (including the 24 MB of L2 cache memory). GPUs can be subdivided into different areas specialising in a particular operation. The major sub-units are the Stream Processors, TMU, and ROP. Newer GPUs are released quicker than CPUs. The major manufacturers of GPUs are NVIDIA (the makers of the GeForce series) and ATI (the makers of the Radeon series).

Here are the features of a few GPUs:

GPUs Compared

GPU	Transistor count (million)	Core Clock	RAM Clock	Memory Bus Width	Stream Processors
GeForce 8800 GT	754	600 MHz	1.8 GHz	256 bit	112
GeForce 8600 GTS	289	675 MHz	2.0 GHz	128 bit	32
Radeon HD 2600 PRO	390	600 MHz	1.0 GHz	128 bit	128
Radeon 3850	666	670 MHz	1.66 GHz	256 bit	320

4.2.16 Graphics Memory

Memory types used as system RAM can also be used for graphics cards in most cases. But high-end cards require memory faster than system RAM. GDDR2 was the first such special-purpose RAM used for graphics cards. The technology behind GDDR2 was different from that which powered DDR2, which was later introduced as